Determinant: Short Notes

How to use these notes: read the Formulas box, follow the Steps and Key points, and avoid the Watch out points.

1. Order and expansion

Formulas

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = a_1(b_2c_3 - b_3c_2) - b_1(a_2c_3 - a_3c_2) + c_1(a_2b_3 - a_3b_2)$$

Sarrus: $D = (a_1b_2c_3 + a_2b_3c_1 + a_3b_1c_2) - (a_3b_2c_1 + a_2b_1c_3 + a_1b_3c_2)$

KEY POINTS

- Order = number of rows = number of columns; a_{ij} lies in row i, column j
- Expand along any row/column; choose the one with most zeros
- Sign pattern for order 3: $+ - + / - + - / + - +$

Watch out

Sarrus rule works only for order 3.

2. Minor and cofactor

Formulas

$$C_{ij} = (-1)^{i+j} M_{ij}$$

$$a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13} = \Delta$$

$$a_{11}C_{21} + a_{12}C_{22} + a_{13}C_{23} = 0$$

$$\begin{vmatrix} C_{11} & C_{12} & C_{13} \\ C_{21} & C_{22} & C_{23} \\ C_{31} & C_{32} & C_{33} \end{vmatrix} = \Delta^2$$

KEY POINTS

- Minor M_{ij}: delete row i and column j; order 3 has 9 minors
- Own row/column $\times$ own cofactors = Δ; other row/column $\times$ these cofactors = 0
- Determinant of cofactors = Δ^{n-1} (order n)

Watch out

Forgetting the sign $(-1)^{i+j}$ when turning a minor into a cofactor.

3. Properties of determinants

Formulas

$$D' = D, \qquad |kA| = k^n|A|$$

$$\begin{vmatrix} a_1 + x & b_1 + y & c_1 + z \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} + \begin{vmatrix} x & y & z \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

KEY POINTS

- Transpose: $D' = D$, so every row result holds for columns
- Swap two rows (columns): sign changes
- Two rows identical or proportional $\Rightarrow D = 0$; a zero row $\Rightarrow D = 0$
- One row $\times k$ gives kD; all rows give $k^n D$
- Add two determinants only if they differ in one row (column)
- $\sum$ goes inside only if one row (column) contains r
- If $D = f(x)$ and $f(a) = 0$, then $(x - a)$ is a factor

Watch out

$|kA| = k^n|A|$, not $k|A|$.
Forgetting that swapping two rows changes the sign.

4. Row and column transformations

Formulas

$R_i \to R_i + \lambda R_j$: D unchanged

$$R_i \leftrightarrow R_j : \ D \to -D, \qquad R_i \to kR_i : \ D \to kD$$
$$R_i \to xR_i + yR_j + zR_k : \ D \to xD$$

STEPS

1. Choose the row/column that creates the most zeros.
2. Apply the operations one after the other, never using an already changed row/column.
3. Take out common factors, then expand along the row/column with zeros.

Watch out

$R_1 \to 2R_1 - R_2$ doubles the value: divide by 2.
$R_1 \to R_1 - R_2$ together with $R_2 \to R_2 - R_1$ is not allowed.

5. Standard determinants

Formulas

$$\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = -(a^3 + b^3 + c^3 - 3abc)$$

$$\begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} = (a-b)(b-c)(c-a)$$

$$\begin{vmatrix} 1 & a & a^3 \\ 1 & b & b^3 \\ 1 & c & c^3 \end{vmatrix} = (a-b)(b-c)(c-a)(a+b+c)$$

$$\begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix} = (a-b)(b-c)(c-a)(ab+bc+ca)$$

$$\begin{vmatrix} 1+a & 1 & 1 \\ 1 & 1+b & 1 \\ 1 & 1 & 1+c \end{vmatrix} = abc + ab + bc + ca$$

$$\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ bc & ca & ab \end{vmatrix} = (a-b)(b-c)(c-a)$$

KEY POINTS

- Look for a row/column of 1's or powers x, x^2, x^3; make it by taking common factors
- Cyclic pattern $(a,b,c),(b,c,a),(c,a,b)$: add all rows to get $a+b+c$

6. Two important operations; AP

STEPS

1. A row/column of 1's: subtract it from the others, expand along it.
2. Rows (columns) add up to a common factor: $R_1 \to R_1 + R_2 + R_3$, take the factor out.
3. Entries of each row in AP: $C_1 \to C_1 - 2C_2 + C_3$ gives a zero column, so $D = 0$.

7. Multiplication of determinants

Formulas

$|A||B| = |AB|$ (row×column; also row×row etc.)

$$\begin{vmatrix} C_{11} & C_{12} & C_{13} \\ C_{21} & C_{22} & C_{23} \\ C_{31} & C_{32} & C_{33} \end{vmatrix} = D^2$$

$$S_r = \alpha^r + \beta^r + \gamma^r : \ |S_{i+j-2}| = [(\alpha-\beta)(\beta-\gamma)(\gamma-\alpha)]^2$$

KEY POINTS

- (i, j) element = sum of products of line i of the first and line j of the second
- Clue: elements are sums of 2-3 products, squares like $(a+b)^2$, or power sums S_r
- Order 3 built from only 2 products per element $\Rightarrow D = 0$

8. Determinant equations, polynomials, trigonometric entries

STEPS

1. To solve $\Delta(x) = 0$: simplify with operations, factor, read the roots; check the count against the degree.
2. Polynomial determinant: constant term $\Delta(0)$, sum of coefficients $\Delta(1)$, degree from the highest power in a product of one element per row and column.
3. Trigonometric entries: use $R_i \to R_i - R_j$ to cancel common parts, reduce to $a + b\sin 2x$, then take range $[a - |b|, a + |b|]$ or solve in the interval.

9. Two variables: lines

Formulas

$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$: unique (intersecting lines)
$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$: infinite (coincident)
$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$: none (parallel)

KEY POINTS

- Three lines, two variables: concurrent $\Leftrightarrow D = 0$ (not all parallel); $D \neq 0$ means no common point
- Three coincident lines give infinitely many solutions

10. Cramer's rule

Formulas

$$x = \frac{D_x}{D}, \quad y = \frac{D_y}{D}, \quad z = \frac{D_z}{D}$$

D_x: replace the x-column of D by d_1, d_2, d_3

KEY POINTS

- Rule applies directly when $D \neq 0$
- For two equations: $D = \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}$ and the same replacement idea

11. Three equations, three variables: all cases

KEY POINTS

- $D \neq 0$: unique solution (three planes meet at a point)
- $D = 0$ and some $D_x, D_y, D_z \neq 0$: no solution
- $D = D_x = D_y = D_z = 0$: infinite or none; solve to decide
- When $D = 0$: find $R_3 = \lambda R_1 + \mu R_2$; consistent iff $d_3 = \lambda d_1 + \mu d_2$
- Infinite solutions: put $z = t$, solve for x, y (planes through a common line)
- Prism: $D = 0$, no two planes parallel, constants violate the row relation

Watch out

Three parallel planes: $D = D_x = D_y = D_z = 0$ yet no solution.

12. Homogeneous system

Formulas

$D \neq 0 \Rightarrow x = y = z = 0$ (trivial only); $D = 0 \Rightarrow$ non-trivial solutions

KEY POINTS

- A homogeneous system is always consistent
- Non-trivial = non-zero = infinite solutions: same condition $D = 0$
- Geometrically: planes through the origin meet in a line

13. Common mistakes checklist

STEPS

1. Write k^n (not k) when all elements are multiplied by k.
2. Correct the value when an operation multiplies the row being changed.
3. Use the cofactor sign $(-1)^{i+j}$.
4. Before saying 'infinite', check the case $D = D_x = D_y = D_z = 0$.
5. For a system with a parameter, substitute the special values back and check each case separately.
6. Do not expand a determinant fully when operations or standard results do it faster.